CS6313- Mini Project 4

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Section 1

1. (12 points) Consider the data stored in the file VOLTAGE.CSV on eLearning. These data come from a Harris Corporation/University of Florida study to determine whether a manufacturing process performed at a remote location can be established locally. Test devices (pilots) were set up at both the remote and the local locations and voltage readings on 30 separate production runs at each location were obtained. In the dataset, the remote and local locations are indicated as 0 and 1, respectively.
(a) (3 points) Perform an exploratory analysis of the data by examining the distributions of the voltage readings at the two locations. Comment on what you see. Do the two distributions seem similar? Justify your answer.

The two distribution does not seem similar. The are differences the shape and center. Remote center produces higher voltage readings on average and is skewed with significant low-voltage outlier. Local location data follows normal, symmetric patterns.
Below are the boxplot and the

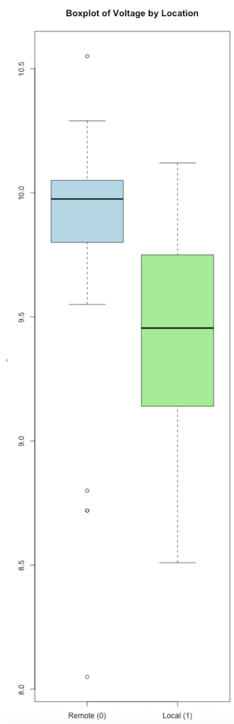

Fig. 1. (a) Boxplot

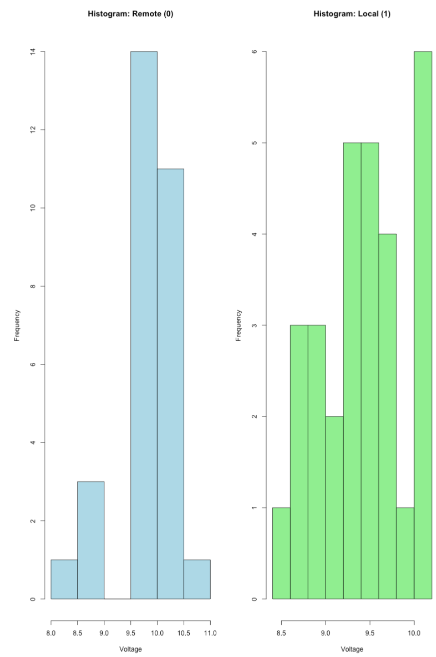

Fig. 2. (b) Histogram

(b) (6 points) The manufacturing process can be established locally if there is no difference in the population means of voltage readings at the two locations. Does it appear that the manufacturing process can be established locally? Answer this question by constructing an appropriate confidence interval. Clearly state the assumptions, if any, you may be making and be sure to verify the assumptions.

```
data:  remote and local
t = 2.8911, df = 58, p-value = 0.005394
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.1173110 0.6453556
```

The goal is to determine if the manufacturing process can be established locally. The criterion is that this is possible if there is no difference in the population means ($\mu_{\text{Remote}} - \mu_{\text{Local}} = 0$). We address this by constructing a 95% confidence interval for the difference in means.

The 95% confidence interval for the difference in population means ($\mu_{Remote} - \mu_{Local}$) is $(0.117, 0.645)$. The CI does not contain zero (it is entirely positive). This means there is a statistically significant difference between the population mean voltages, with the Remote location consistently producing a higher voltage.

(c) (3 point) How does your conclusion in (b) compare with what you expected from the exploratory analysis in (a)?

(b) is consistent with the expectations from the exploratory analysis in part (a).

2. (8 points) The file VAPOR.CSV on eLearning provide data on theoretical (calculated) and experimental values of the vapor pressure for dibenzothiophene, a heterocycloaromatic compound similar to those found in coal tar, at given values of temperature. If the theoretical model for vapor pressure is a good model of reality, the true mean difference between the experimental and calculated values of vapor pressure will be zero. Perform an appropriate analysis of these data to see whether or not this is the case. Be sure to justify all the steps in the analysis.

```
data:  remote and local
t = 2.8911, df = 58, p-value = 0.005394
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.1173110 0.6453556
```

P-value Approach: The p-value of 0.8492 is much greater than the significance level ($\alpha = 0.05$). We therefore fail to reject the null hypothesis (H_0).

Confidence Interval Approach: The 95% CI, (-0.00826, 0.00689), contains zero. This indicates that zero is a plausible value for the true mean difference (μ_d).

Therefore, it does not appear that the manufacturing process can be established locally, as the voltage readings at the two locations are statistically different.

Section 2

Code for Part 1

```r
# Clear workspace
rm(list = ls())

# --- Load the Data ---
# Ensure VOLTAGE.csv is in your working directory
data <- read.csv("VOLTAGE.csv")

# Inspect the data structure
str(data)
head(data)

# Separate the data into two vectors for easier handling
# Location 0 = Remote, Location 1 = Local
remote <- data$voltage[data$location == 0]
local <- data$voltage[data$location == 1]

# Count of samples
n_remote <- length(remote)
n_local <- length(local)

# ============================================================
# Part (a): Exploratory Analysis
# ============================================================

# 1. Summary Statistics
cat("\n--- Summary Statistics: Remote (0) ---\n")
summary(remote)
cat("Standard Deviation:", sd(remote), "\n")

cat("\n--- Summary Statistics: Local (1) ---\n")
summary(local)
cat("Standard Deviation:", sd(local), "\n")

# 2. Visualizations
# Set up a 1x2 plotting area to show Boxplot and Histograms side-by-side
par(mfrow = c(1,1))   # reset to a single plot pane

# Boxplot comparing both locations
boxplot(remote, local,
    names = c("Remote (0)", "Local (1)"),
    main = "Boxplot of Voltage by Location",
    ylab = "Voltage",
    col = c("lightblue", "lightgreen")
)

# Histograms
hist(remote, main = "Histogram: Remote (0)", xlab = "Voltage", col = "lightblue")
hist(local, main = "Histogram: Local (1)", xlab = "Voltage", col = "lightgreen")
```

Code for Part 2

```r
# ============================================================
# Section 2: R Code for Question 2 (VAPOR.CSV Analysis)
# ============================================================

# 1. Load the Data
# Ensure VAPOR.csv is in your working directory
data_vapor <- read.csv("VAPOR.csv")

# Display column names and structure
cat("--- Data Structure ---\n")
str(data_vapor)

# 2. Calculate the Differences (d)
# We define the difference as: d = Experimental - Theoretical
data_vapor$Difference <- data_vapor$experimental - data_vapor$theoretical

# Extract the differences for analysis
differences <- data_vapor$Difference

# 3. Verification of Normality Assumption
# The paired t-test requires the differences to be normally distributed.
# H0: Differences are normally distributed
cat("\n--- Shapiro-Wilk Test for Normality of Differences ---\n")
shapiro_diff <- shapiro.test(differences)
print(shapiro_diff)

# Visualization for Normality (Q-Q Plot)
# A Q-Q plot helps visually assess normality. Points should fall close to the line.
cat("\n--- Q-Q Plot of Differences ---\n")
qqnorm(differences, main = "Q-Q Plot of Vapor Pressure Differences")
qqline(differences)


# 4. Perform Paired Samples t-test
# H0: mu_d = 0 (Mean difference is zero, model is good)
# Ha: mu_d != 0 (Mean difference is not zero, model is not good)
cat("\n--- Paired Samples t-test Results (H0: Mean Difference = 0) ---\n")
t_test_result <- t.test(
  data_vapor$experimental,
  data_vapor$theoretical,
  paired = TRUE,  # Crucial argument for paired t-test
  conf.level = 0.95
)

print(t_test_result)
```